MASON J. HERMAN

(646) 673-6040 · herman.mas@northeastern.edu · linkedin.com/in/masonjherman · github.com/MasonJHerman

EDUCATION

Northeastern University, College of Engineering, Boston, MA *Expected Graduation: June 2029*

B.S., Mechanical Engineering & Bioengineering (Combined Degree), Minor: Artificial Intelligence *GPA: 3.62*

- **Relevant Coursework:** Thermodynamics, Mechanics of Materials, Bioelectricity, Quantitative Physiology, Measurement/Experimentation/Statistics for Bioengineers, Program Design & Implementation I-II

Professional Training: SOLIDWORKS Certified Associate (CSWA) Exam Preparation — In Progress

ORGANIZATIONS

Northeastern Electric Racing • NURobotics VEXU Subdivision • Engineers Without Borders • Black Engineering Student Society • American Society of Mechanical Engineers • Biomedical Engineering Society • NU Barkada (Northeastern Filipino Student Assoc)

TECHNICAL SKILLS

- **Engineering:** SolidWorks (Assemblies, Drawings), AutoCAD, 3D Printing, Soldering, Stress Analysis, SKF Tools
- **Electronics:** Arduino, LTspice, Breadboarding, Oscilloscope, Function Generator, Multimeter, Circuit Analysis
- **Lab & Analysis:** MATLAB, Mechanical Testing, Statistical Analysis, ANOVA, Cell Culture, Fluorescence Microscopy, ImageJ
- **Programming/Tools:** Python, Java, C#, JavaScript, HTML/CSS, Git, GitHub, VS Code, Jupyter Notebook, Notion, Unity

PROJECTS

Personal AI Platform (Independent Project) *June 2026 - Present*

- Developing a modular, privacy-focused AI platform designed for fully local, cloud-independent operation
- Building model-independent architecture to support controlled agent workflows and secure integration with future applications and devices

Powertrain Team, Northeastern Electric Racing (NER) *Fall 2025 - Spring 2026*

- Designed and evaluated powertrain components for an all-electric Formula-style race car, applying statics and SKF tools to assess bearings between the differential motor and hub/gear shaft
- Modeled gear shafts and structural interfaces in SolidWorks, using stress simulations to validate integrity while reducing weight
- Designed mounting joints for electrical storage and major components, considering stability, packaging, and vehicle integration

AI4ALL Program, Stanford University *June 2022 - July 2022*

- Co-developed an image-based feedback controller for autonomous aircraft taxiing in the X-Plane simulator, modeling autonomous navigation.
- Implemented PID control, neural networks, and graph search algorithms to optimize aircraft steering and trajectory tracking
- Collaborated with teammates to train and evaluate computer vision and machine learning models in Python on real-time simulation data, minimizing path deviation

EXPERIENCE

Certified Coding Teacher, Goldeneye Foundation *Sept 2022 - May 2024*

- Delivered weekly one-on-one and small-group Python and Java instruction to under-resourced secondary students in Jamaica, adapting lesson and pacing to each student's skill level

Junior Teacher, TeachForth *Summer 2023*

- Led daily Python instruction for camper cohorts and collaborated with fellow instructors to troubleshoot code and adapt lesson plans in real time

Kitchen Assistant, GlenArbor Golf Club *Summer 2023*

- Prepared food by hand as directed by the sous chef, and cleaned dishes and sanitized equipment and workspaces

PUBLICATIONS

Herman, M., Knezevic, I., Lorenzo, J., et al. (2026, March). *Pulse Ox+: An Exploratory Model for Melanin-Corrected Pulse Oximetry*. Paper presented at the 2026 ASEE Northeast Section Conference, Southern New Hampshire University, NH.

AWARDS

2025: Northeastern Dean's Scholarship, Garnet Baltimore Scholarship Award

2024: Cum Laude Society, AP Scholar with Distinction Award; 2023: National African American Recognition Award